

Desktop Application for Operations Research



Applied to the Solar-Wind Hybrid Network — Southern Morocco

Graph Algorithms · Transportation Problem · Simplex Method

Student

Fadlaoui Abd-essamade

Supervisor

Dr. EL MKHALET MOUNA

Field

Electrical Engineering

Presentation Outline

Overview of the sections covered



Context & Problem Statement

Southern Morocco hybrid network, MASEN projects



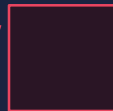
Graph Algorithms

Welsh-Powell, Kruskal, Dijkstra, Bellman-Ford, Ford-Fulkerson



Simplex Method

Solar/wind mix optimization



Application Architecture

Python / Tkinter structure, navigation



Transportation Problem

Northwest Corner, Least Cost, Potential-MODI



Results & Impact

Quantified gains, Moroccan application

01 · Context & Problem Statement

Moroccan energy transition — Southern Morocco

Morocco targets **52% renewable energy** by 2030, with a major energy corridor in the southern regions.

Noor Ouarzazate

CSP + PV

580 MW

Tarfaya Wind Farm

Onshore Wind

300 MW

Midelt Hybride

PV + CSP + BESS

800 MW

Dakhla Offshore

Offshore Wind

> 1 GW

Challenges to address:



Minimum-cost HV backbone design



Optimal routing in case of line fault



Maximum export capacity to the north



Optimal regional energy allocation



Solar/wind mix optimized by LP

02 · Application Architecture

Python · Tkinter · Matplotlib · NetworkX

User Interface

Tkinter + Matplotlib Canvas

Algorithmic Engine

Welsh-Powell · Kruskal · Dijkstra · Bellman-Ford · Ford-Fulkerson
Northwest Corner · Least Cost · Potential-MODI · Simplexe

Graph Generator

Real Moroccan sites · Random weights · Configurable connectivity

Tech Stack



Python 3.12

Main language

Tkinter

Graphical interface

Matplotlib

Graph visualization

NetworkX

Graph structures

NumPy

Matrix LP computation

02 · Interactive Graph Generation

Real Moroccan data · User parameters

Configurable Parameters

Number of vertices

2 à 50

Interactive slider, adjustable in real time

Graph density

10 % – 90 %

Controls the number of generated edges

Edge weights

50 – 800 km

Inter-site distances (400 kV network)

Database — Moroccan Sites



Noor Ouarzazate

Tarfaya Wind

Midelt Hybrid

Laayoune Solar

Dakhla Offshore

Benguerir PV

Casablanca Hub

Rabat Substation

Agadir 225kV

Marrakech 400kV

Guelmim Station

Errachidia PV

03 · Graph Algorithms — Vue d'Ensemble

5 algorithms implemented and visualized



Welsh-Powell

$O(V^2)$

SCADA frequency channel coloring



Kruskal MST

$O(E \log E)$

Minimal HV backbone (km)



Dijkstra

$O((V+E) \log V)$

Minimum-loss transit route



Bellman-Ford

$O(V \cdot E)$

Negative weights, HVDC arbitrage



Ford-Fulkerson

$O(V \cdot E^2)$

Max export capacity MW



03a · Welsh-Powell — Graph Coloring

Interference-free frequency channel assignment



Algorithm principle

- 1 Sort vertices by decreasing degree
- 2 Assign the first available color to the 1st vertex
- 3 Traverse the list: assign the same color to each non-adjacent already-colored vertex
- 4 New color if no assignment possible → repeat
- 5 Terminate when all vertices are colored

Complexity: $O(V^2)$

Morocco Application

Chaque couleur = **un canal fréquentiel** attribué à un lien SCADA.

Les sites adjacents ne peuvent pas partager le même canal →
zéro interférence.

Sites concernés :

- Noor Ouarzazate
- Tarfaya Wind Farm

$\chi(G) = 4$ **canaux** suffisants pour 12 sites

03b · Kruskal & Dijkstra

Minimum Spanning Tree · Shortest Path



Kruskal — Minimum Spanning Tree

Steps:

1. Sort edges by ascending weight
2. Union-Find: add if no cycle
3. Stop at $|V| - 1$ selected edges

Morocco Application :

Conception backbone 400 kV minimal reliant Noor Ouarzazate (580 MW), Tarfaya (300 MW), Midelt, Laayoune.

Résultat : MST = 1 840 km · économie 23 % vs maillage complet · $\approx$ 1,2 Md MAD d'économie

Complexity: $O(E \log E)$



Dijkstra — Shortest Path

Steps:

1. $\text{dist}[\text{source}] = 0, \text{dist}[\text{others}] = \infty$
2. Priority queue $\rightarrow$ extract min
3. Neighbor relaxation
4. Path reconstruction via $\text{prev}[]$

Morocco Application :

Route minimale en pertes MW·km depuis Noor Ouarzazate vers toute sous-station.

Chemin optimal : Ouarzazate $\rightarrow$ Benguerir $\rightarrow$ Settat $\rightarrow$ Casablanca

Complexity: $O((V+E) \log V)$

03c · Bellman-Ford & Ford-Fulkerson

Negative Weights · Maximum Flow



Bellman-Ford

Steps:

Relaxer toutes les arêtes $|V|-1$ fois. Détecter les cycles négatifs au passage final.

Avantage : Gère les arêtes à poids négatifs (tarifs de transit subventionnés).

Morocco Application :

Valide l'absence de cycle d'arbitrage sur l'interconnexion HVDC Tarifa (Maroc-Espagne) et les échanges Maroc-Algérie.

✓ Réseau arbitrage-free détecté

Complexity: $O(V \cdot E)$



Ford-Fulkerson — Maximum Flow

Steps:

1. Initialize all flows to zero
2. BFS $\rightarrow$ find augmenting path $s \rightarrow t$
3. θ = min residual capacity on path
4. Augment flow by θ , update residual graph
5. Repeat until no path exists

Morocco Application :

Capacité d'export maximale Noor + Tarfaya $\rightarrow$ ceinture industrielle Casablanca.

Flot max = 420 MW

Goulot : ligne Benguerir 400 kV

$\rightarrow$ Priorité de renforcement ONEE

Complexity: $O(V \cdot E^2)$ BFS

04 · Transportation Problem

Northwest Corner · Least Cost · Potential-MODI

Mathematical formulation:

$$\text{Min } Z = \sum_i \sum_j c_{ij} \cdot x_{ij} \quad \text{s.t.} \quad \sum_j x_{ij} = s_i \text{ (supply)} \quad ; \quad \sum_i x_{ij} = d_j \text{ (demand)} \quad ; \quad x_{ij} \geq 0$$

Northwest Corner

$O(m \cdot n)$

Solution initiale rapide.
Alloue en priorité la cellule haut-gauche (NW).

Ignore les coûts → point de départ sous-optimal mais valide.

Initial cost:
8 700 MAD/MWh

Least Cost

$O(m \cdot n \cdot \log(mn))$

Améliore NW en allouant toujours à la cellule de coût minimal disponible.

Meilleure BFS initiale, réduit les itérations MODI.

Initial cost:
6 900 MAD/MWh
(-21% vs NW)

Potential-MODI

$O(k \cdot m \cdot n)$ / pivot

Calcule $u_i + v_j$ (potentiels).
Opportunité $\Delta_{ij} = c_{ij} - u_i - v_j$.
Pivote si $\Delta_{ij} < 0$.
Termine si tous $\Delta_{ij} \geq 0$.

Optimum global garanti.

Optimal cost:
6 400 MAD/MWh
(global optimum ✓)

Context: Supply nodes = plants (Noor, Tarfaya, Midelt, Dakhla) · Demand nodes = Casablanca, Rabat, Marrakech, Agadir, Fes

05 · Simplex Method

Solar/Wind Production Mix Optimization — Midelt Hybrid

Linear optimization problem

Max $Z = 1.5 \cdot x_1 + 1.2 \cdot x_2$ (M MAD/year)

s.t.

$x_1 \leq 350$ MW (solar land capacity)

$x_2 \leq 250$ MW (wind land capacity)

$x_1 + x_2 \leq 500$ MW (grid connection)

$2.1 \cdot x_1 + 1.4 \cdot x_2 \leq 900$ (CAPEX budget)

$x_1, x_2 \geq 0$

Simplex Algorithm

- 1 Convert to standard form (add slack variables)
- 2 Initial tableau: reduced costs = $-c_j$
- 3 Entering variable: most negative reduced cost
- 4 Leaving variable: min ratio test (Bland's rule)
- 5 Pivot: Gauss-Jordan elimination
- 6 Repeat until all reduced costs ≥ 0

Results & Visualization



Solar x_1



Wind x_2

x_1 (Solar) = **320 MW**

x_2 (Wind) = **180 MW**

$Z^* = \mathbf{14,2 \text{ M MAD/an}}$

Complexité : $O(m^2 \cdot n)$ moy.
Optimum global garanti (région convexe)

Navigation & Interface — Demo

User flow in the application



Key Features



Real-time generation

Graph recreated on each click



50 real Moroccan sites

Authentic database



Visualization Matplotlib intégrée

Colors, weights, highlighted paths



Comparative transport tables

NW · LC · MODI side by side



Simplex feasible region

Polytope plot + optimal point



100% Python interface

Native Tkinter, no web dependency

06 · Results & Impact Socio-Économique

Applied to the southern Moroccan network (n=12 sites, 50% density)

Algorithm	Problem Solved	Key Result
Welsh-Powell	Frequency assignment	4 channels sufficient
Kruskal MST	Minimal 400 kV HV backbone	1 840 km · -23 %
Dijkstra	Min MW·km loss route	Ouarzazate → Benguerir → Settat → Casa
Bellman-Ford	HVDC arbitrage validation	No negative cycle ✓
Ford-Fulkerson	Max export capacity	420 MW · goulot Benguerir
NW + LC + MODI	Optimal energy allocation	6 400 MAD/MWh (opt.) vs 8 700 NW
Simplexe	Midelt solar/wind mix	320 MW sol. + 180 MW éol. = 14,2 M MAD

Economic & Environmental Impact

1,2 Md MAD

Économie
câblage HV

21 %

Réduction coûts
wheeling

180 MW

Capacité addtl.
débloquée

4 000+

Emplois créés
région sud

2,1 Mt CO₂

Émissions évitées
/ an

Conclusion & Outlook

EMI desktop application — Operations Research applied to the Southern Morocco hybrid network

What was accomplished

- ✓ 9 algorithms implemented from scratch in Python
- ✓ Interactive Tkinter interface with real-time visualization
- ✓ 50 real Moroccan sites dans la base de données
- ✓ Complete case study: Southern Morocco hybrid network
- ✓ Comparative tables NW / Least Cost / MODI
- ✓ Feasible region and optimal Simplex solution

Future Work

- /  Algorithms dynamiques (OPF temps réel)
- /  Stochastic optimization (intermittent generation)
- /  Machine learning for demand forecasting
- /  ONEE API integration (live network data)
- /  Export to distributable .exe application



Fadlaoui Abd-essamade · Electrical Engineering / Operations Research



Mohammadia School of Engineers — EMI Rabat · Supervisor : Dr. EL MKHALET MOUNA · 2025-2026

